

New hips, knees and childbirth: where antibiotic use peaks in Belgian hospitals!

Hospital antimicrobial prescribing in Belgium: linking reimbursement data with clinical context

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Objectives:

- ✓ Identify diagnoses associated with the highest hospital antibiotic use in Belgium
- ✓ Facilitate inter-hospital benchmarking
- ✓ Assess prescribing compliance with clinical guidelines

Methods

AMDIA: AntiMicrobial consumption data linked with **DI**Agno**S**es.

Study Design: Retrospective analysis of 2017 surveillance data from adults in Belgian acute hospitals

Definitions

Antimicrobial consumption:

WHO ATC/DDD methodology Index 2024

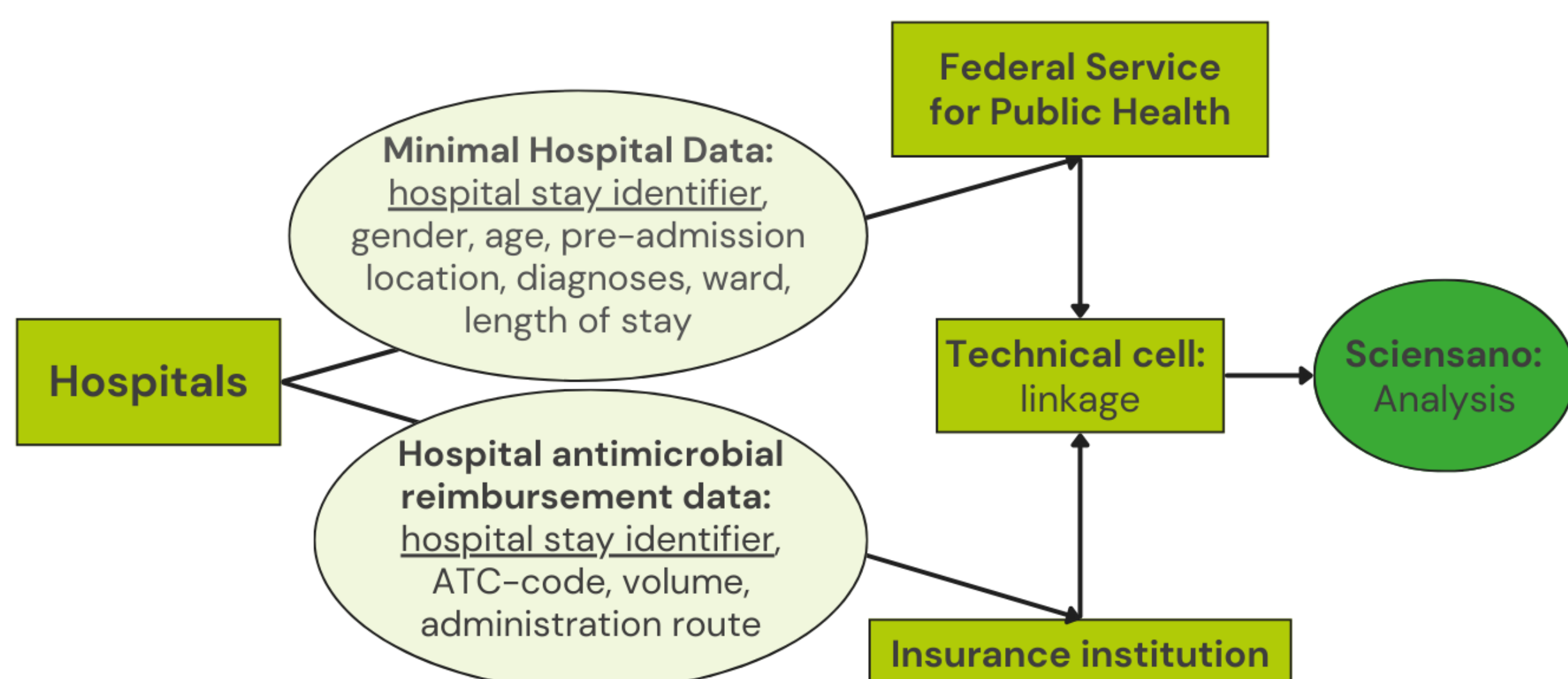
Diagnoses: APR-DRG codes

- All Patient Refined Diagnosis Related Group codes
- Categorize hospital stays by diagnosis, procedures, and illness severity.

Broad-spectrum: piperacillin with enzyme inhibitor (J01CR05), third-generation cephalosporins (J01DD), fourth-generation cephalosporins (J01DE), monobactams (J01DF), carbapenems (J01DH), fluoroquinolones (J01MA), glycopeptides (J01XA), polymyxins (J01XB), and linezolid (J01XX08)

Analyses

- Diagnoses with highest number of hospitalizations involving J01 antibiotics and proportion of hospital stays receiving J01 and broad-spectrum antibiotics
- Diagnoses with highest total J01 DDD in hospitals
- Diagnoses with highest number of hospitalizations involving broad-spectrum antibiotics



Results

	1,6 million hospitalizations	20% cefazolin	of hospital stays
	50% J01 ANTIBIOTICS	8% amoxi-clav	
	13% BROAD SPECTRUM	3% ciprofloxacin	

- Highest number of hospitalizations involving J01 antibiotic consumption in Belgian hospitals in 2017: **Hip and knee replacements and vaginal deliveries**

	30,115 hospitalizations	74% cefazolin	of hospital stays
	98% J01 ANTIBIOTICS	6% rifampicin IV	
	8% BROAD SPECTRUM	3% clindamycin	
	85,272 hospitalizations	17% cefazolin	of hospital stays
	43% J01 ANTIBIOTICS	10% benzylpenicillin	
	0,5% BROAD SPECTRUM	4% ampicillin	
	25,736 hospitalizations	87% cefazolin	of hospital stays
	98% J01 ANTIBIOTICS	4% clindamycin	
	3% BROAD SPECTRUM	2% gentamicin	

- Highest total J01 DDD in hospitals: **pneumonia, chronic obstructive lung disease & septicemia**
- Highest number of hospitalizations with broad-spectrum antibiotics: **pneumonia, chronic obstructive lung disease, kidney and urinary tract infections**

	24,719 hospitalizations	96% J01 ANTIBIOTICS	of hospital stays
	47% BROAD SPECTRUM		
	29,735 hospitalizations	72% J01 ANTIBIOTICS	of hospital stays
	37% BROAD SPECTRUM		
	7,570 hospitalisations	97% J01 ANTIBIOTICS	of hospital stays
	8% BROAD SPECTRUM		
	17,985 hospitalizations	96% J01 ANTIBIOTICS	of hospital stays
	57% BROAD SPECTRUM		

Empowering antimicrobial stewardship with data

- ✓ Priority diagnoses for antimicrobial stewardship in Belgian hospitals
- 🐢 Challenges with delayed (reimbursement) data hinders real-time evaluation of stewardship programs
- 📅 Ongoing efforts for real-time national antimicrobial consumption surveillance system linked with clinical indication and resistance information